

Rubric and Solution of Quiz 3 Part 1

This file contains the **rubric and solution of Quiz 3 Part 1**.

Note;- There may be multiple correct ways to solve a problem. The document presents one representative approach, and the rubric is based on that.

If your solution follows a different valid method with proper justification, appropriate marks will be awarded.

Part 1

Q1

1) 4 marks :-

Solution :- The statement "Every surjective function $f: X$ to Y is injective" is not true in general. Surjectivity, or being "onto," simply guarantees that every element in the codomain Y is mapped to by at least one element from the domain X . This property does not prevent multiple distinct elements in X from mapping to the same single element in Y . Therefore, a function can be surjective without being injective (one-to-one). However, the statement does become true under a specific condition: if X and Y are finite sets that contain the same number of elements (i.e., $|X| = |Y|$). In this special case, the only way to ensure every element in Y is mapped to (surjectivity) given the equal sizes is for each element in X to map to a unique element in Y . This necessity forces the function to also be injective, and thus bijective.

States the correct conclusion regarding the statement - 1 marks

Provides clear and logically valid justification - 3 marks

Or

Counter example to disprove the statement - 2 marks

Proper explanation of counter example - 2 marks

2) 3 marks :-

solution :- Consider the set $\{1, 2, \dots, 100\}$. All odd numbers form a 50-element subset $\{1, 3, 5, \dots, 99\}$, and all even numbers form a 50-element subset $\{2, 4, 6, \dots, 100\}$. Numbers within either of those subsets differ by at least 2, so selecting all 50 odd numbers (or all 50 even numbers) produces 50 selections with no two numbers differing by at most 1. Now suppose we make one more selection, bringing the total to 51. If our first 50 selections were all the odd numbers, the 51st selection must be an even number (because there are no more odd numbers left). Any even number e satisfies either $e-1$ or $e+1$ is odd, and that odd neighbor is already among the 50 odd numbers we chose. Hence the 51st selection creates a pair of numbers that differ by exactly 1.

States a correct minimum guaranteed value - 0.5 marks

Proper use of pigeon Hole - 1.5 marks

Provides logical justification - 1 marks

3) 3 marks :-

Solution:- To prove the set equality

$(A \cap B)' = A' \cup B'$, we must show that each set is a subset of the other.

First, we prove $(A \cap B)' \subseteq A' \cup B'$. Let x be any element such that $x \in (A \cap B)'$.

By the definition of complement, this means $x \notin (A \cap B)$. This logically implies $\neg(x \in A \wedge x \in B)$. By De Morgan's law for logic, this statement is equivalent to $(\neg(x \in A)) \vee (\neg(x \in B))$, which translates to $x \notin A$ or $x \notin B$. By the definition of complement, this means $x \in A'$ or $x \in B'$.

Finally, by the definition of union, this implies $x \in (A' \cup B')$. Thus, we have shown $(A \cap B)' \subseteq A' \cup B'$.

Second, we prove the other direction, $A' \cup B' \subseteq (A \cap B)'$. Let y be any element such that $y \in (A' \cup B')$. By the definition of union, this means $y \in A'$ or $y \in B'$. By the definition of complement, this translates to $y \notin A$ or $y \notin B$. Logically, this is $(\neg(y \in A)) \vee (\neg(y \in B))$.

Applying De Morgan's law for logic, this is equivalent to $\neg(y \in A \wedge y \in B)$.

This means it is not the case that y is in both A and B , so $y \notin (A \cap B)$. By the definition of complement, this implies $y \in (A \cap B)'$. Thus, we have shown $A' \cup B' \subseteq (A \cap B)'$.

Since we have proven inclusion in both directions $(A \cap B)' \subseteq A' \cup B'$ and $A' \cup B' \subseteq (A \cap B)'$, we can conclude that the sets are equal: $(A \cap B)' = A' \cup B'$

Solving $(A \cap B)' \subseteq A' \cup B'$.

- 1.5 marks

Solving $A' \cup B' \subseteq (A \cap B)'$.

- 1.5 marks

Quiz 3 Part 1 Question 1.2 Solutions

Question 1.2.1

Let $F : \mathbb{Q} \rightarrow \mathbb{Q}$ be defined by

$$F(x) = \begin{cases} 2x + 1 & \text{if } x \geq 0, \\ 2x - 1 & \text{if } x < 0. \end{cases}$$

Proof that F is a function

For every $x \in \mathbb{Q}$, exactly one of the conditions $x \geq 0$ or $x < 0$ holds, so a unique rule applies to each x . In both cases, $2x \pm 1 \in \mathbb{Q}$. Hence, for every input there is a unique output in $\mathbb{Q}$, and therefore F is a function.

Injectivity

Suppose $F(x_1) = F(x_2)$.

- If $x_1, x_2 \geq 0$, then $2x_1 + 1 = 2x_2 + 1$, implying $x_1 = x_2$.
- If $x_1, x_2 < 0$, then $2x_1 - 1 = 2x_2 - 1$, implying $x_1 = x_2$.
- If $x_1 \geq 0$ and $x_2 < 0$, then $2x_1 + 1 = 2x_2 - 1$, so $2x_1 - 2x_2 = -2$, thus $x_1 - x_2 = -1$. Since $x_2 < 0$, we obtain $x_1 = x_2 - 1 < -1 < 0$, which contradicts $x_1 \geq 0$.

Hence no such distinct x_1, x_2 exist that map to the same value across different branches, and F is injective.

Not surjective

The range of F is

$$\{y \in \mathbb{Q} : y \geq 1 \text{ or } y < -1\}.$$

Consider $y = 0 \in \mathbb{Q}$. If $F(x) = 0$:

$$2x + 1 = 0 \implies x = -\frac{1}{2}, \quad \text{which contradicts } x \geq 0,$$

$$2x - 1 = 0 \implies x = \frac{1}{2}, \quad \text{which contradicts } x < 0.$$

Thus 0 has no preimage under F , so F is not surjective. Consequently, F is not bijective.

Question 1.2.2

Prove or disprove: $B \subseteq A$ if and only if $\text{pow}(B) \subseteq \text{pow}(A)$.

Proof

($\Rightarrow$) Suppose $B \subseteq A$. Let $X \in \text{pow}(B)$. Then, by definition, $X \subseteq B$. Since $B \subseteq A$, we have $X \subseteq A$. Therefore $X \in \text{pow}(A)$, and so $\text{pow}(B) \subseteq \text{pow}(A)$.

($\Leftarrow$) Suppose $\text{pow}(B) \subseteq \text{pow}(A)$. Since $B \subseteq B$, we have $B \in \text{pow}(B)$. Hence $B \in \text{pow}(A)$, and therefore $B \subseteq A$.

Thus, $B \subseteq A$ if and only if $\text{pow}(B) \subseteq \text{pow}(A)$.

Question 1.2.3

Let $F : A \rightarrow B$ be a function. For any subset $Z \subseteq A$, define

$$F(Z) = \{F(a) \in B : a \in Z\}.$$

Prove: $F(X \cup Y) = F(X) \cup F(Y)$.

Proof

($\subseteq$) Let $b \in F(X \cup Y)$. Then there exists $a \in X \cup Y$ such that $F(a) = b$. Since $a \in X \cup Y$, either $a \in X$ or $a \in Y$. Thus $b \in F(X)$ or $b \in F(Y)$, and therefore $b \in F(X) \cup F(Y)$.

($\supseteq$) Let $b \in F(X) \cup F(Y)$. Then either $b \in F(X)$ or $b \in F(Y)$. If $b \in F(X)$, there exists $a \in X$ with $F(a) = b$. Since $X \subseteq X \cup Y$, we have $b \in F(X \cup Y)$. The argument is identical if $b \in F(Y)$.

Hence $F(X) \cup F(Y) \subseteq F(X \cup Y)$.

Combining both directions yields $F(X \cup Y) = F(X) \cup F(Y)$.

1.2) Question 1:

- Proof F is a Function (1.5 Marks): Uniqueness of image for all x in Q .
- Bijectivity Analysis (2.5 Marks):
 - Injectivity Proof: Show $F(x_1) = F(x_2)$ implies $x_1 = x_2$.
 - Non-Bijective Conclusion.
 - Non-Surjectivity Proof. Correct counterexample (e.g., $y=0$) with proof that it lacks a pre-image.

Question 2:

- One Direction Proof (1.5 Marks): Correct proof for either $B \subseteq A \rightarrow \text{pow}(B) \subseteq \text{pow}(A)$ OR the reverse.
- Both Directions Proof (2.5 Marks): Complete proof of equivalence (iff), showing both directions.

Question 3: (3.5 Marks Total)

- First Inclusion Proof (2.0 Marks): Prove $F(X \cup Y) \subseteq F(X) \cup F(Y)$ by analysing the element a in $X \cup Y$.
- Both Directions Proof (3.5 Marks): Complete proof of equality by also proving $F(X) \cup F(Y) \subseteq F(X \cup Y)$ (i.e., showing $a \in X \Rightarrow a \in X \cup Y$).

Quiz-3 rubric Q1.3

Q1:

We have: A, B, C non-empty sets. $F: A \rightarrow B$, $G: C \rightarrow A$, $H: C \rightarrow A$ functions.

We are to prove or disprove:

$$F \text{ is injective} \iff (\forall G, H: C \rightarrow A, F \circ G = F \circ H \implies G = H).$$

Here $F \circ G = F \circ H$ means: for all $c \in C$ $F(G(c)) = F(H(c))$ $F(G(c)) = F(H(c))$.

Step 1: Proof of ($\implies$)

Assume F is injective.

Let $G, H: C \rightarrow A$ be such that $F \circ G = F \circ H$

That means: $\forall c \in C$ $F(G(c)) = F(H(c))$

By injectivity of F , $G(c) = H(c)$ for all $c \in C$ Thus $G = H$ as functions.

So the implication

$F \circ G = F \circ H \implies G = H$ holds for all G, H

Step 2: Proof of ($\impliedby$)

Assume: For all $G, H: C \rightarrow A$, $F \circ G = F \circ H \implies G = H$

We must show F is injective.

Let $a', a'' \in A$ with $F(a') = F(a'')$ We want to prove $a' = a''$

Pick C to be some non-empty set for simplicity, let $C = \{*\}$ a one-element set.

Then any functions $G, H: C \rightarrow A$ are uniquely determined by $G(*) = a'$, $H(*) = a''$, for some $a', a'' \in A$

The condition $F \circ G = F \circ H$ implies $F(a') = F(a'')$ By our assumption, $F(a') = F(a'')$ forces $G = H$ i.e. $a' = a''$.

Hence, $G = H$ as functions means $G(*) = H(*)$, so $a' = a''$. Thus we conclude:

$\forall a', a'' \in A, F(a') = F(a'') \implies a' = a''$. Therefore, F is injective.

Rubric:

Statement and Understanding (1 mark)

- Clearly restate what is given and what needs to be proved.
- Identify all sets, variables, or functions involved.

Assumptions and Definitions (1 mark)

- State any assumptions or known properties being used.
- Define important terms (like injective etc.) if relevant.

Proof Steps in forward and backward direction(2 marks)

- Use correct logical reasoning or mathematical operations to connect given facts to the conclusion.
- Show each step clearly avoid skipping reasoning.
- Use valid examples or counterexamples if required.

Conclusion and Clarity (1 mark)

- Clearly restate what was proven.
- Summarize how the steps lead to the final result.

If only either forward/backward proof is shown then 2.5 marks will be awarded

Q2: Solution: $\{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}, \{1,2,3\}\}$

- Clearly stating all power sets - **2 marks**
- Any set missing - **1 marks**

Q3:

- Definition of range: $\{x+3: x \in \mathbb{Z}\}$ - **1 mark**
- Showing that for all $y \in \mathbb{Z}$, $x = y - 3 \in \mathbb{Z}$ - **1 mark**
- Concluding $\text{Range}(F) = \mathbb{Z}$, so it's not proper. - **1 mark**